

Given an array of n integers, rearrange them so that the sum of the absolute differences of all adjacent elements is minimized. Then, compute the sum of those absolute differences. Example $n = 5$ $arr = [1, 3, 3, 2, 4]$ If the list is rearranged as $arr' = [1, 2, 3, 3, 4]$, the absolute differences are $|1 - 2| = 1$, $|2 - 3| = 1$, $|3 - 3| = 0$, $|3 - 4| = 1$. The sum of those differences is $1 + 1 + 0 + 1 = 3$. Function Description Complete the function `minDiff` in the editor below. `minDiff` has the following parameter:
 arr: an integer array Returns: int: the sum of the absolute differences of adjacent elements Constraints $2 \leq n \leq 105$ $0 \leq arr[i] \leq 109$, where $0 \leq i < n$ Input Format For Custom Testing The first line of input contains an integer, n , the size of `arr`. Each of the following n lines contains an integer that describes `arr[i]` (where $0 \leq i < n$). Sample Case 0 Sample Input For Custom Testing STDIN Function ---
 ----- 5 $\rightarrow$ `arr[]` size $n = 5$ 5 $\rightarrow$ `arr[] = [5, 1, 3, 7, 3]` 1 3 7 3 Sample Output 6 Explanation $n = 5$ $arr = [5, 1, 3, 7, 3]$ If `arr` is rearranged as $arr' = [1, 3, 3, 5, 7]$, the differences are minimized. The final answer is $|1 - 3| + |3 - 3| + |3 - 5| + |5 - 7| = 6$. Sample Case 1 Sample Input For Custom Testing STDIN Function ---

1, $|2 - 3| = 1$, $|3 - 3| = 0$, $|3 - 4| = 1$. The sum of those differences is $1 + 1 + 0 + 1 = 3$. Function

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 ----- 2 $\rightarrow$ arr[] size $n = 2$ 3 $\rightarrow$ arr[] = [3, 2] 2 Sample Output 1 Explanation $n = 2$ arr = [3, 2] There is no need to rearrange because there are only two elements. The final answer is $|3 - 2| = 1$.

Answer: (penalty regime: 0 %)

Reset answer

```
5  * The function accepts INTEGER_ARRAY arr as parameter .
6  */
7
8  int minDiff(int arr_count, int* arr)
9  {
10     arr.sort()
11     total_dif =0
12     for i in range(1,len(arr));
13     total_dif+=abs(arr[i] -arr[i-1])
14     return total_dif
15
16 }
17
```

You are transporting some boxes through a tunnel, where each box is a parallelepiped, and is characterized by its length, width and height.

The height of the tunnel **41** feet and the width can be assumed to be infinite. A box can be carried through the tunnel only if its height is strictly less than the tunnel's height. Find the volume of each box that can be successfully transported to the other end of the tunnel. Note: Boxes cannot be rotated.

Input Format

The first line contains a single integer **n** , denoting the number of boxes.

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Vo 5G 96%

```
2 int main()
3 {
4     n= int(input())
5     boxes =[]
6     for_in range(n;
7     volume =box_volume(n,boxes)
8     print(volume)
9
10 }
```

9:51

Vo 5G 96%



Week-14-Structures and Unions: Attempt review | REC-CIS
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You are given n triangles, specifically, their sides a_i , b_i and c_i . Print them in the same style but sorted by their areas from the smallest one to the largest one. It is guaranteed that all the areas are different.

The best way to calculate a volume of the triangle with sides a , b and c is Heron's formula:

$$S = \sqrt{p * (p - a) * (p - b) * (p - c)} \text{ where } p = (a + b + c) / 2.$$

Answer: (penalty regime: 0 %)

```
1 #include<stdio.h>
2 int main()
3 {
4     for in range(n);
5     for trinagle in triangles;
6     print(a,b,c)
7 }
```


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Vo 5G 96%



Week-15-Pointers: Attempt review | REC-CIS
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Given an array of integers, reverse the given array in place using an index and loop rather than a built-in function.

Example

`arr = [1, 3, 2, 4, 5]`

Return the array `[5, 4, 2, 3, 1]` which is the reverse of the input array.

Function Description

Complete the function `reverseArray` in the editor below.

`reverseArray` has the following parameter(s):


```
28 *      *(a + i) = i + 1;
29 *  }
30 *
31 *      return a;
32 *  }
33 *
34 */
35 int* reverseArray(int arr_count, int *arr, int *result_count) {
36     start = 0
37     end = len(arr)-1
38     while start < end;
39     start += 1
40     end -= 1
41     return arr
42
43 }
44
```

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Week-15-Pointers: Attempt review | REC-CIS
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An automated cutting machine is used to cut rods into segments. The cutting machine can only hold a rod of *minLength* or more, and it can only make one cut at a time. Given the array *lengths[]* representing the desired lengths of each segment, determine if it is possible to make the necessary cuts using this machine. The rod is marked into lengths already, in the order given.

Example

$n = 3$

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Week-15-Pointers: Attempt review | REC-CIS

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```
1  /*
2  * Complete the 'cutThemAll' function below.
3  *
4  * The function is expected to return a STRING.
5  * The function accepts following parameters:
6  * 1. LONG_INTEGER_ARRAY lengths
7  * 2. LONG_INTEGER minLength
8  */
9
10 /*
11 * To return the string from the function, you should either do static a
12 *
13 * For example,
14 * char* return_string_using_static_allocation() {
```

```
21 *   char* s = malloc(100 * sizeof(char));
22 *
23 *   s = "dynamic allocation of string";
24 *
25 *   return s;
26 * }
27 *
28 */
29 char* cutThemAll(int lengths_count, long *lengths, long minLength) {
30     lengths I[4,3,2,5]
31     rod_length =20
32     minlength =3
33
34
35 }
36
```